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ལས་ཁུངས།

མངའ་སྡེ་རྣམས་ཚལ་ཡིག་ཚང་། བཀའ་བསྐྱེད་།

རྣམས་ཚལ་སྤྱིང་འཛིན་ཡིག་ཚང་། བཀའ་གཡང་སྤྱིང་།



Cross-cutting Management Regime (NWFP) for

**Bimkhar Bamboo Management Group
(1st Revision)**



Plan Period: January 2025 - December 2035

Yangste Geog

Trashiyangste Dzongkha

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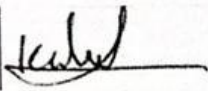
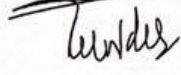
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DoFPS/FRPMD/FRPIS- NWFP-01/2025/925 (March 31, 2025)

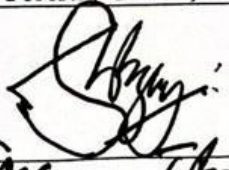
Approval Sheet

Name of the NWFP Management and Marketing Group	Bimkhar Bamboo Management Group
Dzongkhag	Trashigang
Gewog	Yangtse
Village(s)	Bimkhar
No. of households	48
Area (ha)	234.9
Plan Period	January 2025 – December 2034

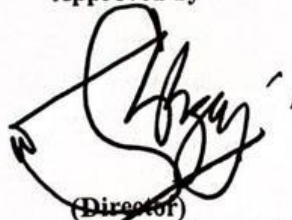
Prepared and submitted for approval:

	Chairperson of NWFP Management and Marketing group	Gup, Gewog Administration	Head, FRPMS, Division Office	Chief Forestry Officer, Division Office
Signature				
Name	Gonpo Tshering	Goenpo	Tenzin Wangdi	Karma Leki
Title	Chairperson	Gup	DCFO	CFO

Reviewed and recommended for approval:

	Reviewing Officer, FRPMD	Chief Forestry Officer, FRPMD
Signature		
Name	Kezang Nangmo	Samten Tenzin

Approved by



(Director)

Department of Forests and Park Services



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PART A: BIMKHAR BAMBOO MANAGEMENT AND MARKETING PLAN

Background

The Bimkhar Bamboo cross cutting management regimes is prepared for sustainable management and marketing of non-wood forest products. The Bimkhar Bamboo Cross-Cutting Management Regime (234.9 hectares) completely falls inside the area-based management plan of Biological Corridor No.9 under the jurisdiction of Divisional Forest Office, Trashigang and is prepared for implementation by the local communities of Yangtse Gewog.

Bimkhar (273626.61N & 912950.7E) is one of the closest Chiwogs of Yangtse Gewog among five Chiwogs within the Gewog under Trashi Yangste Dzongkhag. The village is located in the northeast direction from Gewog center (273626.61N and 912950.7 E) at an elevation of 1,700m. All the villages under the Chiwog have access to farm roads. The community practices agricultural and livestock farming in addition to wood curving or other off-farm activities for sustainability of their livelihoods. They also depend on forest resources for timber, firewood, fodder, leaf litter, non-wood forest products (NWFPs) and other basic needs.

The Bimkhar Bamboo Management and Marketing Group (BBMMG) comprises 48 household members from six nearby villages namely Goenpadag, Kelmang, Tsherigom, Chukchi, Tongchen and Bimkhar. The group is entrusted to govern the net production areas of 234.9 hectares of bamboo forest in State Reserved Forest (SRF) land. The bamboo management area lies towards the northeast direction of the Chiwog, and it takes 1 to 3 hours walking distance from the settlement. It is stratified into four major blocks namely, Zulusep_Shanakhar with 86.5 ha, Relamserula_Rotsha with 78 ha, Jangpeng with 41.3 ha, and the Zhabaymo block with 29.1 ha.

Additionally, the management areas fall inside the landscape of Biological Corridor Nine (BC9). The vegetation composition in management area is dominated by temperate cool broadleaved forest with tree species such as *Betula alnoides*, *Quercus spp*, *Acer spp*, *Michelia spp*, etc, and marginally conifer forest with *Pinus bhutanica* (Bhutan white pine) and *Tsuga dumosa* (hemlock). The understory vegetation comprises *Borinda grossa*, other small bamboo species, *Daphne spp.*, and associated fern species. With elevation ranging from 2,600m to 3,200 m, the management area has a slope from 1 percent to 45 percent.

Bimkhar Bamboo Management and Marketing Group was initially established in 2020 with funding assistance by the World Wildlife Fund (WWF) through the Department of Forests and Park Services, and the Trashigang Forest Division. Whereas the Yangtse Forest Range had assisted the technical backstopping for consultation, resource assessment, plan and by-laws preparation. Subsequently, in 2020, its five-year management plan was officially approved by the Department of Forests and Park Services.

With the expiry of the last management plan, a revised management plan is being drawn for another 10 years term. The current management plan and by-laws are developed in accordance with the provisions of the Forest and Nature Conservation Rules & Regulations of Bhutan, 2023 and the Forest and Nature Conservation Code of Best Management practices of Bhutan Volume V: Cross-Cutting management regimes. The revision was done based on the benefits ripped from the earlier plan period, and the group members' willingness to continually govern the management area.

The plan revision has been carried out jointly by the group and the Yangtse Forest Range from plan revision consultation meeting to resources assessment, management plan and by-laws preparation. Additionally, Yangtse Range has consistently facilitated technical backstopping in incorporating better scopes and strategies, where both management of resources and marketing of their products are being prioritized, and planned for effective implementation in the 10 years plan period.

Review and achievements of Past Management Plan Period (2020-2024)

Physical achievements

- The sale of bamboo products (either raw materials or finished products) is an asset in enhancing the livelihood and boosting the economy for the local communities of Bimkhar chiwog.
- The formation of this Bimkhar bamboo management group facilitated in bringing strong collaboration among 48 households of the community consisting of six villages.
- Upon knowing all the harvesting procedures and norms, the group harvested bamboo following a proper scientific and sustainable approach. This had not just helped them earn a living for themselves but also kept provisions for the younger generations to benefit in the future.

Financial achievements

Table 1. Financial achievement of the group in the past management plan period (2020-2024)

Period	Products	Unit	Quantity harvested	Month of harvest	Total Income (Nu.)
2020-2024	Bamboo raw materials	Pieces	22087	December to March	20277.69
<i>Twenty Thousand Two Hundred Seventy-Seven and Sixty-Nine Chhetrum</i>					

Observation and findings

- The community needs to reinforce good governance and explore ways to diversify income streams.
- The group needs to strengthen maintenance of proper and transparent documentation practices.
- The group needs to initiate timely patrol and field assessment inside managed areas during resource harvesting period to curb illegal extraction of the bamboo resources.

Recommendations

- The formation of the group had not just benefited the local community at large but also been able to protect bamboo and other forest resources from illegal activities, catchment areas and wildlife conservation, thus, the continuity of this group is strongly recommended.
- To strengthen good governance, group management or administrative work and transparent maintenance of recording keeping or documentation, therefore, the time to time need base and up-scale training for executive or committee members is crucial.
- The group members require training to have indigenous knowledge and exceptional skills for producing or diversifying various bamboo products to improve their income generation, hence, the product diversification training is highly recommended as requested by the group members with letter no. nil. dated 22/09/2024
- Explore strategies to improve fund accumulation and support loans among members.

Objectives

This management and marketing plan aims to achieve the following objectives:

- To sustainably manage bamboo resources and ensure the fair distribution of resources through effective management.
- To promote and diversify bamboo products by applying traditional knowledge and skills.
- To ensure economic development, which will contribute to the poverty alleviation and income generation activities in the Bimkhar communities.
- To protect catchment areas of the forested area.

Description of Bamboo

There are many species of small bamboos in Yangtse Gewog of which only two species namely *Borinda grossa* and *Chimonobambusa callosa* (Rashuu in Sharchopkha) are usually collected by the people for fencing, temporary shed roofing and flagpole purposes. Moreover, *Borinda grossa* is also used in weaving baskets and other essential items. However, only *Borinda grossa* is collected from Bimkhar bamboo management area.

Table 2. Brief description of *Borinda grossa* for rural and commercial purposes

Name	Harvesting area	Part used	Use purpose	Harvesting season	Estimated quantity that can be collected
<i>Borinda grossa</i>	All four blocks	Culms	Mats, baskets, fencing, roofing & flag poles for rural & commercial	December to February	5000 pics

Prescriptions for sustainable harvesting of *Borinda grossa*

The management and marketing plan for this bamboo group is to protect the *Borinda grossa* bamboo from over harvesting. This is further to ensure maintain its' sustainable harvesting from the designated area, and prescriptions are described based on traditional, scientific and Framework for Management and Marketing of NWFPs, 2021. The prescriptions shall be strictly adhered by the group members while harvesting bamboo resources from the four blocks.

Scientific harvesting system

Table 3. Harvesting prescriptions for sustainable management of resources

NWFP species	Harvesting prescriptions for sustainable harvesting
<i>Borinda grossa</i>	Culms should be cut at the base or at first long internodes above node
	Bamboo shoots or culms of less than one-year old should be left intact or to retain
	A maximum of 33% of the 1+ year and 2+ year old culm should be harvested each year and 67% should be kept for regeneration. This Annual Harvesting Limited (AHL) can be applied to each available clump
	70% of the 3+years old culms can be harvested. It is important to leave some old culms to provide physical support to the 1+year and 2+years old culms.
	Culms should be harvested when new shoots are sprouting as this practice will harm the new developing shoots. The preference timing for harvesting bamboo culms is four to five years after the bamboo- shooting seasons and at least one to two months before (This season differs according to location and altitude)
	Respect the traditional belief (<i>la-dam</i> and <i>Ri-dam</i>) related to the allowed timing for entering the forest which overlaps with the bamboo-shooting seasons.
	Harvesting of condensed/congested clumps should be done in a U-form or horseshoe form to maximize culms in a U-form that will allow more shoots to grow.

Boundary demarcation of the bamboo management area

The boundary demarcation of the Bimkhar bamboo management area was done through consultation with group members. It was mapped based on the traditional boundaries, their customary practices, and the availability of preferable *Borinda grossa* bamboo resources. It has an area of 234.9 hectares inside BC 9, with no private registered land. The boundary of the bamboo management area is described in table 3 and figure 1. However, the group lacked the management and harvesting rights over timber and other forest products.

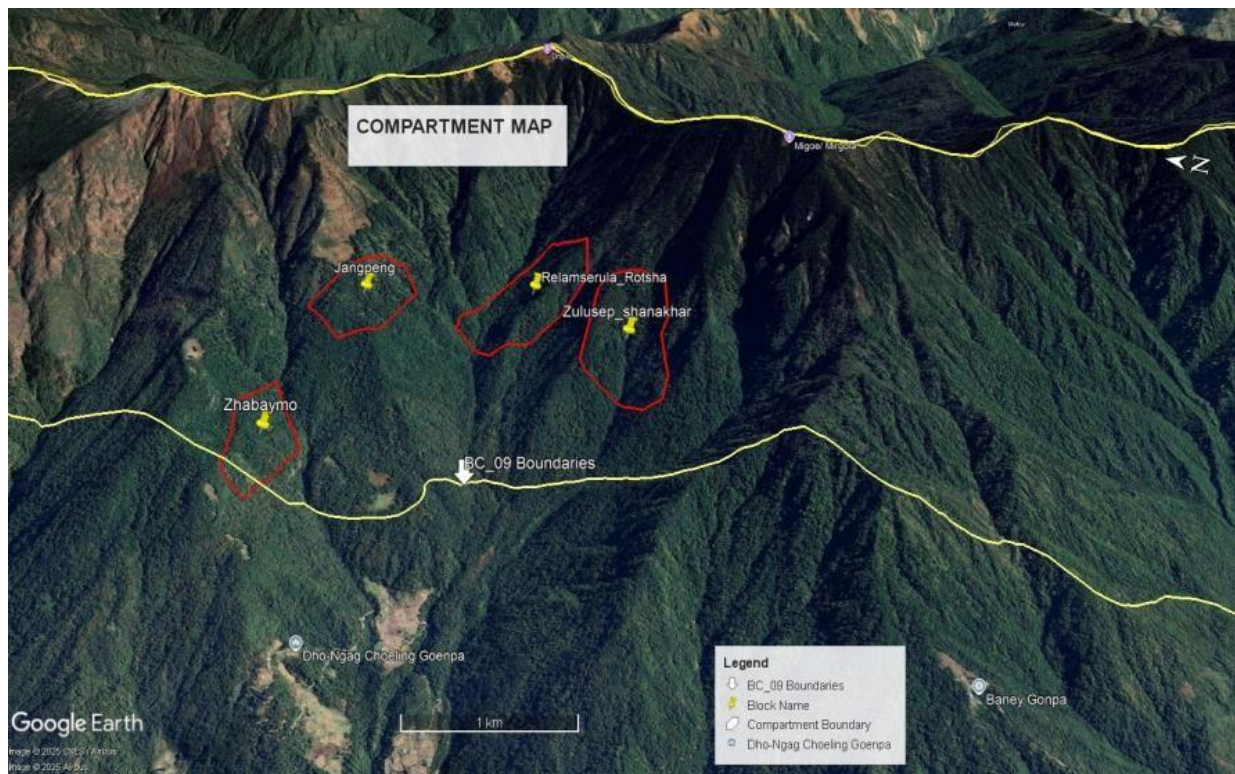


Figure 1. Google map showing the overall boundary of the Bimkhar Bamboo Management area

Table 4. Details of boundary of bamboo management area

Boundaries	Description of the permanent features
East	Dribla Base
South	Tsheringom Top
West	Zulusep
North	Zhubaymo

Resource Assessment

The detailed sketch map with the location of the bamboo management area is shown in figure 2. Additionally, the sketch map of the production or harvesting areas indicated by local names against each compartment is also shown in figure 3. In this, the inaccessible areas were excluded from the total production area measurement. These production areas lie within 1 to 3 hours walking distance from the settlement.

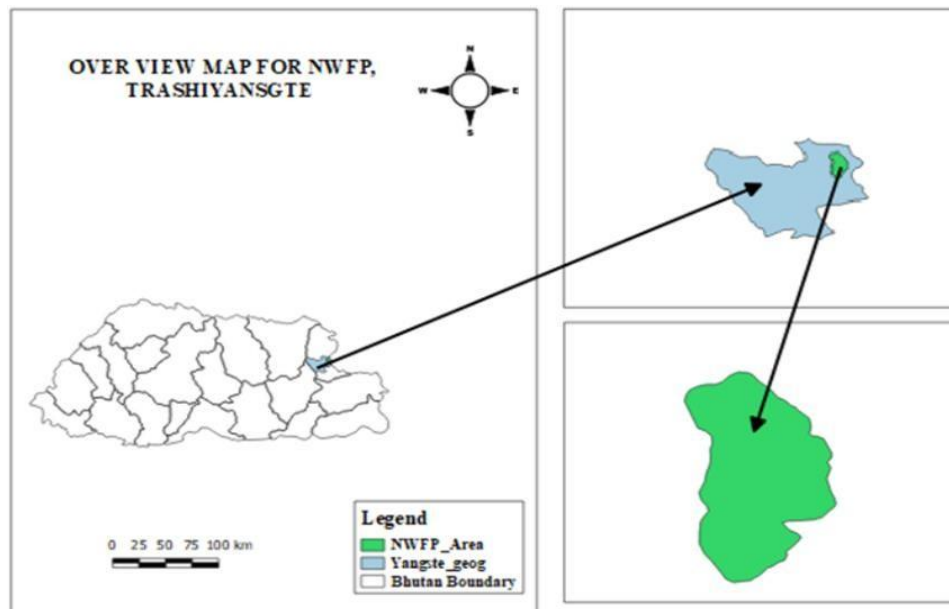


Figure 2. Map showing the overview location of Bimkhar Bamboo Management Area

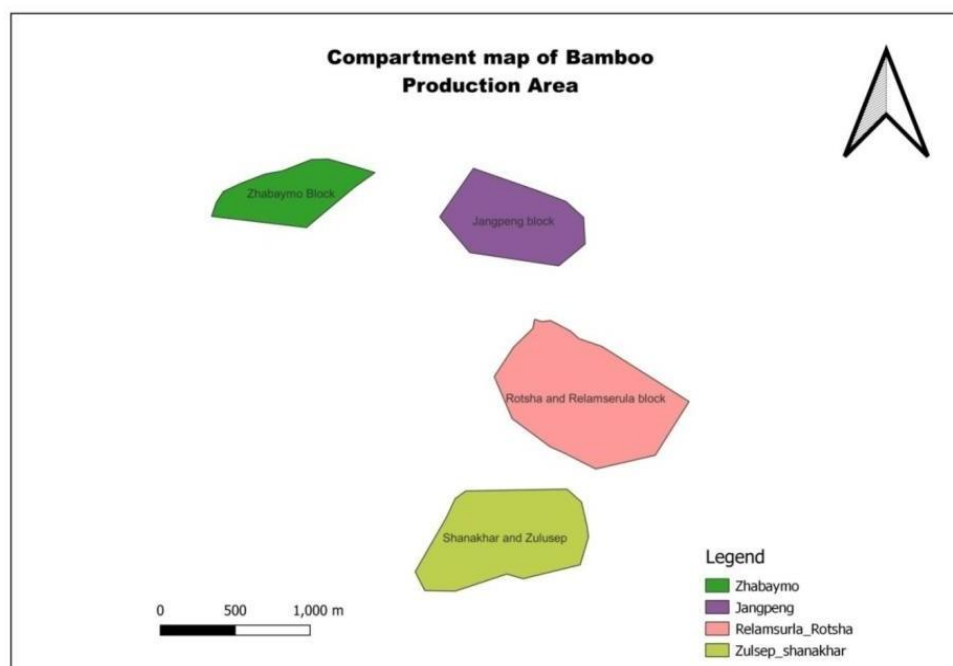


Figure 3. Map showing compartment wise production area of the Bimkhar Bamboo Management Area

Bamboo resource assessment

Bamboo resource assessment is done through participatory mapping for demarcation of boundary which is then necessary for actual mapping and resources assessment. This was done through presentation and consultations with group members who know the exact overall boundary location considering their traditional rights. During the process, extra care had been taken to avoid conflict over resource collection boundaries with neighboring communities.

Four major blocks have been divided based on the availability density and quantity of bamboo resources. Local area names were also given to each block by the group members (figure 3). Similarly, sub-block names were also designated because of the non-existence of local names for this (figure 4). The overall mapping and block division was done using google earth and QGIS tool, and it's being presented to the group members for final authentication.

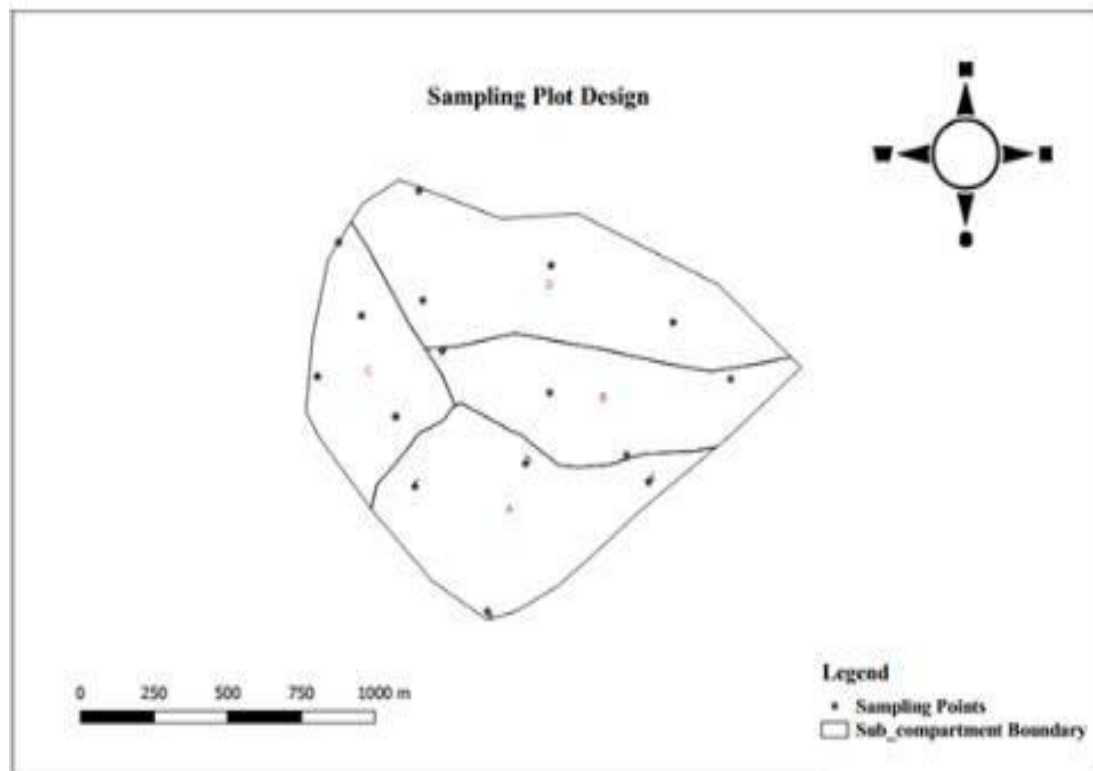


Figure 4. Map showing sample plot layout in a block

The field resource assessment and measurement of bamboo clumps and culms had been carried out by laying down a quadrant of 25mx10m. For measurement of this plot size, a slope correction was also made. A quadrant was selected randomly using “Ran function in excel”. The detailed data collection was based on Guidelines for resources assessment and management for *Borinda grossa* (2008) developed by the Department of Forest and Park Services. Likewise, the data compilation and analysis were also done as per the standard format.

Marketing and sales plan

Table 5. Marketing and sale plan for the group

Name of NWFP	Potential market	Estimated qty. for sale per year (piece)	Expected price (Nu.) per piece	Total value (Nu.) of the expected sale
<i>Borinda grossa</i>	Domestic market	2000	1	2000

The potential markets for the sales of bamboo raw materials and products were properly worked out. Therefore, the important aspect of this plan is to have well planned and strategic marketing arrangements. The harvesting and marketing of bamboo products, both raw materials and finished products, will be organized and arranged by the groups led by the executive committee members. Additionally, equal shares of profits shall be made after deduction of both marketing expenses and contribution to the group saving fund.

Marketing arrangement

The group executive committee members headed by Chairman or coordinator shall oversee and coordinate for the overall marketing of the bamboo raw materials and products.

Table 6. Procedures for harvesting, marketing and sharing of income

Activity	Procedures
Harvesting of products	All the group members shall participate during the harvest and collection of raw materials.
Practices	The postharvest practices such as product diversification in terms of bamboo, and preparation of raw materials
Marketing of products	The group shall appoint individuals from the members who shall carry out marketing. However, the group executive committee members shall oversee and coordinate for the overall marketing of the products in consultation with the members. Product marketing can be done using any social media platforms.
Income sharing	The profit from the sales of the raw materials and products will be shared equally among the group members. This will be divided after having contributed certain share to the group saving fund and clearing the various expenses made for the marketing

Benefits to the members

The profit-sharing mechanism is such that 15% of revenue generated through sales of products (after deducting various expenses) would be deposited to the group saving fund while the remaining 85% of the revenue will be divided amongst the group members. This will help in

boosting the revenue of the group as well as help prospering the individual member. Table 10 shows the financial projection of the benefits to the group members.

Table 7. Financial projection of the benefit to the members through sale of NWFPs

Sl. No	Particulars	Product (names)	Total Amount (Nu)
		<i>Borinda grossa</i>	
1	Revenues		
a	Total quantity (unit)	16230	
b	Market price (Nu per unit)	Nu.1 per pic	8287/-
c	Total revenues from forest product sales (a*b)		8,287/-
A	Total revenues		22,087/-
2	Expenses		
a	Packaging materials		0
b	Royalty		0
c	Transportation cost		0
d	Compensation fee to executive committee for marketing products		0
e	Contribution to community development fund		8287/- (15% of total sale)
f	Incidental expenses		0
B	Total expenses		0
C	Total benefit (net-income) of group members (total revenues A-total expenses B)		22,087/-
D	Average income to group members (C/no of group members)		460.2/-
E	Price per unit paid to group members (C/1a)		172.6/-

Benefits to the group

Natural resources management and marketing mainly focus on generating income for the group and for the individual members but keeping in mind the sustainable techniques are being followed while harvesting of resources, this ensures the sustainability of the resources for the next 10 years, and future generation. The group strategized the profit-sharing mechanism, whereby the 15% of revenue generated through sales of products (after deducting various expenses) would be deposited into the group fund while the remaining 80% would be divided among the group members. Table 11 shows the financial projection of the benefits to the group.

Table 8. Financial projection of the benefit to the group, the community development fund

Sl. No	Particulars	Amount (Nu)
	Revenue	
1	Membership fees	13,800/-
2	Contribution to the community development fund (CDF) from sales of the product 1	8287/-
3	Contribution to the CDF from sales of product 2	00
4	Contribution to the CDF from sales of product 3	00
A	Total revenues	22,087/-
	Expenses (only include expenses that will be paid from the group funds)	6000/-
B	Total Expenses	6,000/-
C	Balance: Total revenues-total expenses	16,087/-

PART 2: GROUP BY-LAWS

The Bimkhar Bamboo Management and Marketing Group (BBMMG) is governed by the 48 household members from six nearby villages of Bimkhar Chiwog under Yangtse Geog, TrashiYangtse.

Table 9. Details of the NWFP groups

Village	Number of members Household	Gewog
Bimkhar, Gonpada, Tshergom, Chukchi, Kelmang, Tongchen	48	Yangtse

Membership Arrangements and Resignation

Membership Agreement	Responsibilities and rights of members: ● A household of Bimkhar community who has “Gung” and “Thram” is eligible to join the group as a member ● Members leaving the group with valid reasons (e.g. resettlement program, illness, etc.) will be paid his/her benefits accumulated till that day, but members leaving on his own interest will not be given any benefits from the group. ● Finances and wages for the meeting and work carried out shall be included. ● Those households who have not joined the group during the initiation of the group formation and wanted to join the group later (after the formal approval of the group) can only become a member after depositing total expenditure right from work execution, other necessary fees and remittance the membership fee and the individual share accumulated till that day.
Member of the Bimkhar Bamboo Management & Marketing Group (BBMMG)	Responsibilities and rights of members: ● Should inform committee members in case of illegal collection within the designated management area by outsiders or by the members. ● Strictly adhere to the harvesting guidelines while harvesting bamboo resources from the culms. ● Patrol the management area during the harvesting season. ● Terminate committee members if their performance is not satisfactory. ● Appoint a new committee member if he/she is found capable. ● Shall contribute labor as and when required for group activities or meetings. ● has the right to conduct auditing on financial management if found guilty. ● I shall attend timely meetings and work. ● Assist executive members in resolving conflicts within the group. ● We shall contribute the annual membership fee and compensation for committee members. ● Collaborate with group Executive Committee Members in regulating FNCRR, Management and Marketing plan rules and regulations, and By-Law. ● Involve in all the activities as per the annual work plan.

Executive committee

	Name of Chairperson: Gonpo Tshering (17 364 067)
Chairperson	<i>Responsibilities of Coordinator/Chairperson:</i> ● Chair the group meetings and activities. ● Submit an annual progress report. ● Attend the visitors if they visit the group. ● Resolve disputes and conflicts within the group. ● Inform the group members of the date for harvesting. ● To coordinate the sale of collected forest products to the traders. ● Make a visit to the sometimes-demarcated area and see the boundary. ● Maintain administrative records. ● Fix schedule for meeting and collection of resources. ● Countersigns for group financial transactions. ● Seek technical backstopping for the group. ● Sign group related correspondence. ● Impose fines and penalties in consultation with group members.

	Name of Secretary: Tshewang Peldon (17 437 877)
Secretary	<i>Responsibilities of Secretary:</i> Officiate as chairperson during his absence. Keep all the important records with regard to the group. Maintain minutes of the meetings. Assist the chairman in terms of group related works. Representative of the group towards the outsider. Assist Chairman and accountant in maintaining records. Maintain records of the group activities carried out by the members. Countersign in financial transactions of the group. Deal with any type of correspondence that is related to the group. Arrange and provide good hospitality for the visitors.

	Name of Treasurer: Tashi Gyeltshen (77230290)
Treasurer	<i>Responsibilities of Treasurer:</i> ● Maintain proper financial records. ● Collect membership fees from the members. ● Take care of financial arrangements. ● Present financial status during the general meeting. ● Countersign in financial transactions. ● Collect fines and royalties. ● Issuing of permits

<i>Tenure of committee member</i>	• The committee members elected from the group members, by the group members unanimously, will serve for 3 years to provide equal opportunity to all members. • However, no executive committee members performing them duties as expected by group members will be terminated before the expiry of the tenure, and the vacant post shall be filled immediately.
<i>Membership fee and purpose</i>	• A sum of Nu. 300/- (three hundred) each would be collected from all the members as a membership fee every three years. The membership fee amount will be deposited in the group's savings account maintained with Bhutan Development Bank Limited (BDBL). The funds collected will be used for the benefit of group members.
<i>Incentives to the committee members</i>	• Since the group is in its infancy stage, there are still needs to accumulate as much funds as possible. Therefore, the executive committee members will be given Nu.2000 (two thousand) only as incentive remuneration every three years.

Group fund management and utilization

The group has a joint group saving account; therefore, all the funds of the group are kept in the group saving account maintained with BDBL. The executive members will have the authority to deposit or withdraw funds/budget by maintaining clear records and documentations. The ceiling amount to be withdrawn from the group account for office and administrative expenses must be decided by the executive committee in consultation with the group members.

<i>Community development fund and utilization of fund</i>	• To improve the economic condition of the group members, the loan will be sanctioned to the members by assessing and confirming genuine needs of the applicants. • The executive member shall consult with the group members about the loan allotment. • Loans shall be sanctioned only to the group members and no loan shall be supported to the outsider. • Loan amount should not exceed 50% of the total fund of the group. • Loan out with interest of 8% per annum or annually. • The loan term shall not exceed one year and if borrower repayment capacity is poor, the term for extension will be decided by the executive member. • The borrower shall pay the principal amount including the interest on an instalment basis annually and upon liquidity of loan, fresh loan can be sanctioned. • If the borrower fails to repay in each time, the penalty of 24% will
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	be imposed per annum and if genuinely had financial disadvantage, the executive committee will decide and reschedule term not more than three months. Annually BBMMG executive committee shall review all the loan repayments and share/present information during the BBMMG general meetings.
<i>Fund transaction procedure</i>	- Open a joint group saving account in BDBL in the name of the group with the signatures of three executive committee members (Chairman, Secretary and Treasurer). - Mandatory requirement of counter sign during the withdrawal of cash from the group saving account from the three executive members. - Any cash generated for the group shall be deposited into the group saving account by the Treasurer.
<i>Conflict resolution</i>	- In case any dispute or conflict arises among members within the group the committee members headed by Chairman will try to solve. If committee members can't solve it, the matter will be forwarded to the Geog Tshogchung (GT) and Dzongkhag Tshogdue (DT) for further deliberation and decision. - In case there is a conflict between the BBMMG and outsiders, the case shall be forwarded to the nearest forest office and Geog Administration by the coordinator (group chairman).
<i>Meeting and purpose</i>	1. Committee meeting - The committee meetings shall be called as and when required. 2. General meeting - The general meetings will be conducted 2 times on a half yearly basis. This is generally to discuss the effective implementation of annual work plan activities, resource allotment and collection for group members, and marketing of bamboo raw materials and products to the outsider. - It shall also discuss issues, and the amount of product harvested, revenue collection, and the achievement of planned activities.
<i>Penalties</i>	- A fine of Nu.1000/- per head load will be imposed other than BBMBG for illegal collection of bamboo products from the designated management area. It shall confiscate the product and sell at fair market price upon clear decision by the committee members. - The fine of Nu.500/- per head load will be imposed within BBMMG

	for their illegal collection of bamboo products. It shall be confiscated and sold at a fair market price upon clear decision by the committee members. • Double fine imposition to the repeated offender be it BBMMG or outsider. • Absentee in meetings should be liable/impose a fine of Nu. 500/- per day or occasion. The group members agree that members should reach the meeting venue 10 minutes before or exactly during the start of the meeting session. • Group work absentees should be imposed a fine of Nu. 600/- per day. • Individual selling of bamboo products is strictly prohibited. The product should sell only through a group. • However, if found guilty a fine should be imposed which will be decided by the management committee. This shall further be guided by FNCA and FNCRR, 2023. • The group members agree to strictly adhere to a sustainable scientific harvesting system and traditional harvesting practice. If any member tries to and violates, it shall not be permitted to collect bamboo resources for one year from the management area.
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Responsibility of Gewog Administration, Yangtse

- Support guidance in conflict resolution and management that are forwarded to the Gewog Tshogdu by the chairman.
- Support budget for group capacity development and product diversification.

Responsibilities of Yangtse Range Office, Trashigang Forest Division

- Facilitate the formation of NWFP group, and support in plan revision.
- Facilitate boundary survey, resource assessments and framing by-Laws.
- Support technical backstopping for the preparation and implementation of the management plans, and AWP activities.
- Facilitate in record keeping, capacity development of the group members and group governance.
- Monitor and verify the records maintained by the group committee members.
- Facilitate and support in conflict resolution and management.
- Support group in marketing bamboo products.
- Prepare annual monitoring report and submission to the Divisional Forest Office, Trashigang.
- Coordinate a midterm review and final evaluation of the implementation of the plan.

Powers of the DoFPS

The department has the right to terminate the BBMMG in case of any violation of the management plan prescription or in the event of inability to carry out the activities as per the plan.

PART 3: MONITORING AND EVALUATION

Monitoring by Group lead by Committee members

Monitoring will be conducted annually but if there are genuine requirements to increase the frequency of the monitoring, it should be done based on the decision of the committee members. The main motive of the monitoring will be to patrol whole blocks collecting information regarding resource management status, threats (disease) to bamboo and illegal activities. Right after the monitoring, the team shall present the report to the group members and come up with a way forward with solutions to resolve. Encountering any illegal activities, the executive members should handle the case seriously as per penalties prescribed in the by-law. It is very serious that the group or coordinator should report immediately to the Yangtse Forest Range office to sort out.

Joint Monitoring by Group and Yangtse Forest Range

The annual monitoring will be carried out jointly by the group/group executive committee members and Yangtse Forest Range. This is mainly to assess the achievement of AWP activities, implementation of management plan and by-law, financial performance, record keeping, and identify key challenges and issues faced by the group. In brief, the annual monitoring generally audits the plan activities falls under the three thematic areas, firstly the NWFP management, secondly the NWFP marketing and fund utilization, and thirdly the group governance. Furthermore, the monitoring team shall carry out physical verification of the respective harvesting blocks inside the designated management area to verify resource availability and assess whether the group has adhered to management plan harvesting prescription. The Forest Range office shall prepare an annual monitoring report and shall submit it to the Divisional Forest Office, Trashigang for record and further submission to the DoFPS.

Review and Evaluation

The midterm review in the fifth year and the final evaluation in the 9th year shall be conducted by the Forest Monitoring and Information Division (FMID), DoFPS.

Table 10. Financial projection for three years for the bamboo group

Sl. No	Particulars	Unit	Rate (Nu.)	Amount (Nu)	Total (Nu.)
1	Membership fee	48 heads	300/-	300/-	14400/-
2	Royalty	Nu.	0.08/-	1200/-	96/-
3	Permit fee	Nu.	10/-	1200/-	12000/-

Table 12. Bamboo Culms Record Form

Name of village			Dzongkhag										
Gewog				Year									
SN	Date	Name of Household	Name of collection areas	Number of Harvested culms									
				1st year	2nd year	3rd year	4th year	5th year	6th year	7th year	8th year	9th year	10th year

Annexure I Resource Assessment Data**1. Resources assessment for Bamboo of Jangpey Block, Bimkhar**

Age Class	Number of Culms	Sub-Block A	Sub-Block B	Total of culms in Block
		Culms	Culms	
1+years old Bamboo culms	Total culms counted in Sample plot	63	108	171
	Total harvestable culms in sample plot	31	55	86
	Total in sub-blocks	54197	96001	150198
	Total harvestable culms in sub-blocks	1280	2271	3551
	AHL (33%)	17885	31680	49565
2+years old Bamboo culms	Total culms counted in Sample plot	54	94	148
	Total harvestable culms in sample plot	27	45	72
	Total in sub-Blocks	47168	78799	125967
	Total harvestable culms in sub-blocks	1115	1858	2973
	AHL (33%)	15565	26003	41568
3+Years old Bamboo culms	Total culms counted in Sample plot	67	85	152
	Total harvestable culms in sample plot	25	32	57
	Total in sub-Blocks	45409	56770	102179
	Total harvestable culms in sub-blocks	1032	1321	2353
	AHL (70% annual harvest limit)	31786	39739	71525

2. Resources assessment compilation sheet for Zhabaymo Block, Bimkhar

Age Class	Number of Culms	Sub-Block A	Sub-Block B	Sub-Block C	Total of culms in Block
		Culms	Culms	culms	
1+ Year old Bamboo culms	Total culms counted in Sample plot	36	39	2	77
	Total harvestable culms in sample plot	16	19	2	37
	Total in sub-blocks	14593.65	17224.5	1751.82	33569.97
	Total harvestable culms in sub-blocks	465.5	552.9	58.2	1076.6
	AHL (33%)	4895.9	5684	578.1	11158
2+years old Bamboo culms	Total culms counted in Sample plot	90	31	6	127
	Total harvestable culms in sample plot	63	28	4	95
	Total in sub-Blocks	55968.03	24612.78	3561.8	84142.61
	Total harvestable culms in sub-blocks	1833.3	814.8	116.4	2764.5
	AHL (33%)	18469.45	8122.21	1171.41	27763.07
3+ Years old Bamboo culms	Total culms counted in Sample plot	593	155	59	807
	Total harvestable culms in sample plot	472	41	35	548
	Total in sub-Blocks	403215.42	39229.71	30336.75	472781.88
	Total harvestable culms in sub-blocks	13735.2	1193.1	1018.5	15946.8
	AHL (70% annual harvest limit)	282250	27460	21235	330945

3. Resources assessment compilation sheet for Zulusep_Shanakhar Block, Bimkhar

Age Class	Number of Culms	Sub-Block A	Sub-Block B	Sub-Block C	Total of culms in Block
		Culms	Culms	culms	
1+ Year old Bamboo culms	Total culms counted in Sample plot	141	164	181	486
	Total harvestable culms in sample plot	67	79	81	277
	Total in sub-Blocks	513507	598450	614712	1726669

	Total harvestable culms in sub-blocks	5795	6833	7006	19634
	AHL (33%)	169457	197488	202855	569800
2+years old Bamboo culms	Total culms counted in Sample plot	102	100	109	311
	Total harvestable culms in sample plot	48	46	52	146
	Total in sub-Blocks	363819	348854	394007	1106680
	Total harvestable culms in sub-blocks	4152	3979	4498	12629
	AHL (33%)	120060	115121	130022	365203
3+Years old Bamboo culms	Total culms counted in Sample plot	47	90	152	289
	Total harvestable culms in sample plot	17	35	54	106
	Total in sub-Blocks	129793	266636	412518	808947
	Total harvestable culms in sub-blocks	1470	3027	4671	9168
	AHL (70% annual harvest limit)	90855	186645	288762	566262

4. Resources assessment for Bamboo of Relamserula_Rotsha Block, Bimkhar

Age Class	Number of Culms	Sub-Block A	Sub-Block B	Sub-Block C	Sub-Block D	Total of culms in Block
		Culms	Culms	Culms	culms	
1+ Year old Bamboo culms	Total culms counted in Sample plot	34	29	38	31	132
	Total harvestable culms in sample plot	11	9	21	18	59
	Total in sub-Blocks	68718	55926	129090	110526	364260
	Total harvestable culms in sub-blocks	858	702	1638	1404	4602
	AHL (33%)	22676	18455	42599	36473	120203
2+years old Bamboo culms	Total culms counted in Sample plot	60	52	62	49	223
	Total harvestable culms in sample plot	19	25	32	24	100
	Total in sub-blocks	118794	154206	197028	147966	617994
	Total harvestable culms in sub-blocks	1482	1950	2496	1872	7800

	AHL (33%)	39202	50887	65019	48828	203936
3+Years old Bamboo culms	Total culms counted in Sample plot	162	190	263	159	774
	Total harvestable culms in sample plot	56	72	120	70	318
	Total in sub-Blocks	348972	447252	741624	432822	1970670
	Total harvestable culms in sub-block	4368	5616	9360	5460	24804
	AHL (70% annual harvest limit)	244280	313076	519136	302975	1379467